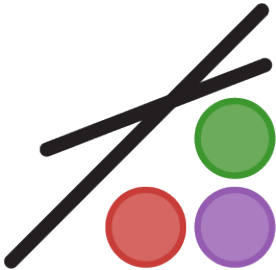


JuMP and MultiObjectiveAlgorithms



version 1.30.0 and more



version 1.12.0 and more



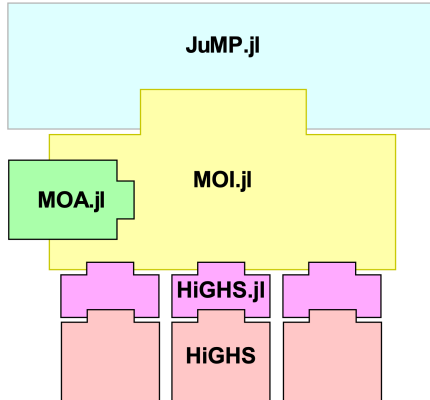
Installing the package:

```
julia> using Pkg
```

```
julia> Pkg.status()  
Status `~/ .julia/environments/v1.11/Project.toml`  
 [60bf3e95] HiGHS v1.18.1  
 [4076af6c] JuMP v1.26.0
```

```
julia> Pkg.add("MultiObjectiveAlgorithms")
```

Getting Started (con't)



Algorithms currently available (v1.11.0)

- | | |
|----------------------------------|------------|
| 1. MOA.Lexicographic() [default] | $p \geq 2$ |
| 2. MOA.Dichotomy() | $p = 2$ |
| 3. MOA.EpsilonConstraint() | $p = 2$ |
| 4. MOA.Hierarchical() | $p \geq 2$ |
| 5. MOA.Chalmet() | $p = 2$ |
| 6. MOA.KirlikSayin() | $p \geq 2$ |
| 7. MOA.DominguezRios() | $p \geq 2$ |
| 8. MOA.TambyVanderpooten() | $p \geq 2$ |
| 9. MOA.GeneralDichotomy() | $p \geq 2$ |

+ optimization attributes coming with a given algorithm

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+ optimization attributes coming with a given algorithm

Zoom on the ϵ -constraint algorithm

Algorithm:

► `EpsilonConstraint()`

Y.V. Haimes, L.S. Lasdon, D.A. Wismer (1971). On a bicriterion formation of the problems of integrated system identification and system optimization. *IEEE Transactions on Systems, Man and Cybernetics*, Volume SMC-1, Issue 3, Pages 296-297.

Attributes:

► `MOA.EpsilonConstraintStep()`

algorithm uses this value as the ϵ by which it partitions the first-objective's space. The default is 1, so that for a pure integer program this algorithm will enumerate Y_N .

► `MOA.SolutionLimit()`

if this attribute is set then, instead of using the `MOA.EpsilonConstraintStep`, with a slight abuse of notation, `EpsilonConstraint` divides the range of the first-objective's domain in objective space by `SolutionLimit` to obtain the ϵ to use when iterating. Thus, at most `SolutionLimit` solutions are returned.